Functional Stability Theory — Physical Instantiation via the Renormalized Free Energy Principle

[RFEP v1.10 maintenance candidate — tangential certificate-ledger guardrail]

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Abstract

Functional Stability Theory (FST) is a programme that identifies a single structural pattern—functional positivity under a gauge constraint—as the common substrate of open problems in number theory, mathematical physics, and cosmology. This paper develops the physical instantiation of FST. We present the Renormalized Free Energy Principle (RFEP) as an organising stability meta-rule that structures Pattern A (PA1–PA3) together with the Dissipative Selection Principle (DS1–DS3).

Three ontological layers are distinguished: (1) the abstract RFEP as an organising stability meta-rule; (2) the Dissipative Selection Principle as the mechanism by which physical evolution selects a dissipation-minimising attractor under explicit hypotheses; and (3) domain-specific instantiations as verification programmes rather than completed domain proofs. We map the framework across five domains: FST-Physics (Yang–Mills, Navier–Stokes, Navier–Stokes Log-Distance Inequality, Turbulence K41) and FST-Cosmology (Dark Energy via the Cooperative Renormalization Model). The historical origin in the Riemann Hypothesis and the mathematical foundation of FST are developed in the companion paper [2]. The paper is provided in English and German versions. This maintenance update also introduces a tangential certificate-ledger reading of RFEP transfers: residual/gap evidence remains diagnostic unless target choice, source-axis independence, outer gap, Galerkin exactness or no-pollution status, projected/tangential residual, paid versus unpaid normal residual, scale degree, advice budget, and matched controls are separately recorded.

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1 Introduction and Problem Statement

1.1 The spectrum-zero identification

Hilbert and Pólya proposed that RH would follow from the existence of a self-adjoint operator H on some Hilbert space $\mathcal{H}$ whose spectrum is the set of imaginary parts of the non-trivial zeros of the Riemann zeta function:

$$\operatorname{spec}(H) = \{\gamma_\rho \in \mathbb{R} : \zeta(1/2 + i\gamma_\rho) = 0\}. \tag{1}$$

For Dirichlet: find H_χ with $\operatorname{spec}(H_\chi) = \{\gamma_\chi^{(k)} : L(1/2 + i\gamma_\chi^{(k)}, \chi) = 0\}$.

This *spectrum-zero identification* is the open step in the Connes programme [9, 10, 11].

1.2 Two obstacles on the direct Hilbert side

1. **Eigenvector analyticity barrier.** For the Dirichlet Weil operator on $L^2([-L, L])$, Paley-Wiener extension is bounded by $\sigma = 1/2$ (archimedean pole, prime-Dirichlet convergence) [6, 7]. The L-zeros do not enter at this level.

2. **NE-B group-structure obstruction.** The prime-shift algebra on $L^2([-L, L])$ has centraliser dimension 1 [5], Section 5. The Semigroup-Group-Equivalence conjecture (SGE; [1], Section 9) predicts that direct Hilbert-Pólya fails without a group structure.

The combination motivates a non-Hilbert, non-direct-Galerkin realisation. Meyer [8] realises this via an idele-class representation on a Fréchet distribution space. FST proposes an axiomatic vocabulary in which Meyer's construction appears as a natural abelian instantiation and in which alternative routes (CCM prolate, Bost-Connes thermodynamic) can be compared.

1.3 Contribution

This v1.10 maintenance candidate reorganises the paper around RFEP as the physical instantiation of Functional Stability Theory. Its contribution is fivefold:

1. it separates the ontological layers RFEP, DS1–DS3, and domain-specific instantiation;
2. it records the post-Zookeeper transfer obligations as explicit operator/form, defect, Galerkin-faithfulness, complement-gap, and trace-compatibility checks;
3. it adds the v1.10 tangential-provenance guardrail, separating total residual, projected tangential residual, paid normal residual, and unpaid escape directions before any RFEP-transfer signal is read as evidence;
4. it tabulates the current verification status across the physics and cosmology branches; and
5. it keeps the original spectrum-zero/RH material branch-local, referring the mathematical foundation to the companion Zeta Zoo paper [2].

No new theorem about RH, Yang–Mills, Navier–Stokes, or cosmology is claimed. The main contribution is a status-disciplined RFEP layer: it distinguishes proved structural reductions, conditional transfers, numerical proof-of-life evidence, and open analytic obligations.

1.4 Relation to prior versions

This is the RFEP v1.10 maintenance candidate. It incorporates the v0.8 post-Zookeeper transfer package, but changes the paper-level role: RFEP is now presented as the physical instantiation of FST rather than as an appendix to the spectrum-zero route map. Relative to the public v1.9 release, this v1.10 candidate revision makes explicit the Tangential-Provenance-Ledger rule from June 2026: a small raw residual/gap value is not certificate evidence unless the target, source axis, projection, tangent gap, normal-rest bookkeeping, and matched controls are all visible. Relative to v0.8, the update adds branch-locality, the Zeta Zoo separation, the physics/cosmology proof-status table, and the CRM circle in the outlook. Relative to v0.7 it retains the Three-Lemma Endgame and the RFEP transfer obligations for physical supplements. Relative to v0.6, claims about Theorem 8.4 (MS1 $\Leftrightarrow$ MS2), Conjecture 8.6 (Weyl law for QW_λ), Proposition 8.2 (CCM instantiates (FST-7)(B)), the trace-class claim for G_β^{ren}, and the residual-gap rhetoric in the Emergence section remain downgraded to match the technical rigour actually present in the supporting notes. An external audit (April 2026) identified systematic overstatements in v0.6; the present version keeps paper-level statements aligned with note-level diagnoses.

2 FST Self-Duality Without Hilbert Space

2.1 What replaces self-adjointness?

Self-adjointness on a Hilbert space enforces real-valued spectrum and complete eigenvector families simultaneously. On a non-Hilbert space, these must be enforced separately.

Definition 2.1 (FST self-duality)**.** A triple $(\mathcal{V}, B, G)$ is an *FST self-dual system* if:

1. $\mathcal{V}$ is a locally convex topological vector space (typically Fréchet or reflexive Banach).
2. $B : \mathcal{V} \times \mathcal{V} \to \mathbb{C}$ is a continuous non-degenerate sesquilinear form (the *self-duality pairing*).
3. G is a locally compact group acting continuously on $\mathcal{V}$ and preserving B.

When $\mathcal{V}$ is Hilbert and B the inner product, this is a unitary G-representation.

Proposition 2.2 (Real spectrum via B-symmetry)**.** *Let $(\mathcal{V}, B, G)$ be FST self-dual. If H is the infinitesimal generator of a one-parameter subgroup acting B-symmetrically ($B(Hv, w) = B(v, Hw)$ on a dense domain), then the joint eigenvalues of H with respect to B are real.*

The argument is standard for B-symmetric operators on reflexive spaces [8].

2.2 Why FST motivates non-Hilbert

FST stability (RFEP; [1], Section 6.5): a system is stable iff its RG-flow converges to a self-similar fixed point.

For an operator family $\{A_\lambda\}_{\lambda>0}$, stability under RG rescaling is not generally compatible with a Hilbert structure. A Fréchet or locally convex space admitting the rescaling as an automorphism is natural. The idele class group is the prototypical example.

3 FST Axioms Formalised

3.1 Ground data

An *FST structure* consists of:

(G) A locally compact abelian topological group G with Haar measure.

(R) A closed subgroup $R \leq G$, isomorphic to $\mathbb{R}$ or $\mathbb{R}_+^\times$ (the *renormalisation axis*).

(N) A complementary closed subgroup $N \leq G$ with $G \cong R \times N$, N compact-profinite (the *arithmetic nucleus*).

(χ) A unitary character $\chi : G \to \mathbb{C}^\times$.

(S) A G-equivariant Bruhat-Schwartz class $\mathcal{S}(G)$, with rapid decay along R, locally constant regularity along N, and a Fourier transform $\mathcal{F} : \mathcal{S}(G) \to \mathcal{S}(\hat{G})$ with Pontryagin duality $\hat{G} \cong G^*$.

3.2 Pattern-A axioms

(P1)
$G = R \times N$ topologically.

(P2)
$\mathcal{S}(G) = \mathcal{S}(R) \otimes \mathcal{S}(N)$ topologically.

(P3)
$\mathcal{F} = \mathcal{F}_R \otimes \mathcal{F}_N$.

3.3 The SGE axiom

(SGE)
G is a group (not a semigroup). Translations $T_g : \mathcal{S}(G) \to \mathcal{S}(G)$ are invertible.

This is the G-analogue of what fails on $L^2([-L, L])$ under prime shifts (cf. NE-B, [5], Section 5).

3.4 The DS1–DS3 axioms

(DS1) ***Dissipation.***
The R-action $\{T_r\}$ is $\mathcal{S}$-preserving, contractive in a weighted seminorm.

(DS2) ***Selection.***
$P_\chi : \mathcal{S}(G) \to \mathcal{S}(G)^\chi$ is continuous, idempotent.

(DS3) ***Reproduction.***
For $N = \prod'_\ell N_\ell$, $\mathcal{S}(G)^\chi = \mathcal{S}(R) \otimes \bigotimes_\ell \mathcal{S}(N_\ell)^{\chi_\ell}$.

3.5 The RFEP axiom

(RFEP)
There exists a free-energy functional $\mathcal{F}_\chi : \mathcal{S}(G)^\chi \to \mathbb{R}$, convex and coercive under the R-action, whose critical points are the R-generator eigendistributions.

These six axioms constitute the *FST baseline*.

Remark 3.1 (Status of the axiom package)**.** As a classification language the axiom package is useful: it cleanly separates a baseline framework from the completeness principle (FST-7) and from later stronger selectors (FST-Free, FST-Min) introduced only in Section 13. As a minimal axiomatics, however, it is *not yet* convincing: minimality and independence are claimed but not proved. Some items (e.g. Pattern A) overlap substantially with the ground data; RFEP is treated as a central principle but its force relative to the other axioms is not sharply isolated; and the later extensions (FST-Free, FST-Min) function as *selection principles encoding arithmetic content* rather than as consequences. The appropriate reading is therefore: the axioms are a useful map of where content lives, not yet a foundational axiom system with proved independence.

3.6 Canonical L-function

For $f \in \mathcal{S}(G)^\chi$,

$$\tilde{f}(s) := \int_R f(r, e_N)\, r^s \,\frac{dr}{r},$$
$$\Lambda_{G,\chi}(s) := Z_\infty(s) \cdot \prod_\ell L_\ell(s, \chi_\ell).$$

This is Tate's thesis [21] axiomatised.

4 Branch-Locality and the Zeta Zoo

The RFEP v1.10 candidate uses a branch-local reading of FST. The mathematical branch grew historically from the Riemann Hypothesis, Hilbert–Pólya programmes, and the Zeta Zoo. That branch supplies the strongest formal vocabulary: group structure, self-duality, gauge constraints, positivity forms, and explicit trace obstructions. It is not, however, the same object as the physical RFEP layer.

In this paper, RFEP is the physical meta-rule:

$$\text{stable physical evolution} \quad \Longleftrightarrow \quad \text{renormalised free energy reaches a constrained minimum.}$$

The Dissipative Selection Principle (DS1–DS3) is the mechanism by which the RFEP minimum is selected dynamically. Domain papers then instantiate this mechanism with concrete functionals, operators, and defect structures.

The Zeta Zoo companion paper [2] is therefore treated as the mathematical foundation and historical origin of FST, not as a prerequisite proof for the physical claims made here. This separation prevents a common over-reading: physical RFEP applications may inherit vocabulary and proof obligations from the Zeta Zoo, but they do not inherit an RH claim.

4.1 Current domain status

Domain	RFEP status	Present proof status
Yang–Mills	Strong-coupling/local-transfer RFEP instance	Volume-independent local/cylinder-sector transfer gap for fixed gauge-invariant local observables; full time-slice L^2 operator gap and continuum mass-gap transfer remain separate gates.
Navier–Stokes	Conditional RFEP/defect instance	L^2 obstruction theorem and numerical reach diagnostics present; global regularity remains conditional on Assumption G and Condition D.
NS Log-Distance Inequality	RFEP-compatible LDI/TLL bridge	BV chain rule proven; PDE-level TLL/LDI closure remains open.
Turbulence K41	Static variational RFEP positive control	Reference-normalised local KL/free-energy minimiser in the reduced shell ansatz; global Talagrand/LSI, UCU3, and dynamical cascade bridges remain open.
Dark Energy / CRM	Cosmological RFEP branch placement	CRM/FST-DE supplies a UCU/screening/model-identification framework; DESI/CPL, observable-window, Cassini/screening, and parameter-stability gates remain open.

Table 1: Branch-local proof status for the RFEP v1.10 candidate revision.

5 The FST Fixed-Point Theorem

5.1 The construction

$\mathcal{V}_\chi$ Quotient of $\mathcal{S}(G)^\chi$ by coinvariants $\mathcal{K}$ (functions with vanishing Tate integrals on the critical line).

B_χ The Tate pairing $B_\chi(v, w) = \int_G v(g)\mathcal{F}[w](g)\chi(g)dg$.

H_χ Generator of R-action.

5.2 The theorem

Theorem 5.1 (FST fixed-point theorem, conditional on (FST-7))**.** *Let $(G, R, N, \chi, \mathcal{S})$ be an FST structure satisfying the baseline axioms and the completeness axiom (FST-7) of Section 7. Then:*
(a) $\mathcal{V}_\chi$ is a non-trivial Fréchet space.
(b) B_χ is non-degenerate, R-equivariant.
(c) H_χ is B_χ-symmetric.
(d) $\mathrm{spec}_{\mathrm{pt}}(H_\chi) = \{\gamma \in \mathbb{R} : \Lambda_{G,\chi}(1/2 + i\gamma) = 0\}$.
(e) $v_\gamma \in \mathcal{V}_\chi^*$ satisfies $B_\chi(v, v_\gamma) = \tilde{v}(1/2 + i\gamma)$.

Corollary 5.2 (abstract RH, conditional)**.** *Under the hypotheses* including (FST-7)*, the non-trivial zeros of $\Lambda_{G,\chi}$ lie on* $\mathrm{Re}(s) = 1/2$.

Parts (a)–(c), (e) follow from Tate's thesis in the adelic case. Part (d) — the spectral identification — requires (FST-7), and this is precisely the input that is not derivable from the baseline axioms. The statement is therefore a *reformulation* of the classical Meyer/Tate content in FST language, not a new theorem.

6 Minimal Example: Riemann Zeta

The minimal non-trivial instantiation: $G = \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$, $\chi = \chi_0$, $R = \mathbb{R}_{+}^{\times}$, $N = \hat{\mathbb{Z}}^{\times}$. The adelic Tate data supply standard instances of the baseline group, character, test-function, and product-formula inputs. The RFEP/DS reading and the (FST-7) completeness condition remain additional structural imports, not trivial consequences of the adelic setup.

6.1 Tate integrals

With $f = f_\infty \otimes \prod_p f_p$, $f_\infty(x) = e^{-\pi x^2}$, $f_p = \mathbf{1}_{\mathbb{Z}_p}$:

$$Z_\infty(s) = \pi^{-s/2}\Gamma(s/2), \qquad Z_p(s) = (1 - p^{-s})^{-1},$$

$$Z(f, s) = \pi^{-s/2}\Gamma(s/2)\,\zeta(s) = \Lambda(s).$$

Adelic self-duality $\hat{f} = f$ gives $\Lambda(s) = \Lambda(1 - s)$ directly.

6.2 The operator

$H_{\chi_0} = -x_\infty \partial_{x_\infty}$, multiplication by $-s$ on the Mellin side. Mellin evaluations $v_\gamma : f \mapsto Z(f, 1/2+i\gamma)$ sit in $\mathcal{V}_{\chi_0}^{*}$. Non-triviality at zeros is (FST-7).

7 The Completeness Axiom (FST-7): Three Views

7.1 (FST-7) algebraic: Pontryagin self-duality

Definition 7.1 (abstract Meyer condition)**.** (FST-7A): there exists a discrete subgroup $\Xi \leq G$ with: *(i)* $\Xi^\perp = \Xi$ under Pontryagin pairing; *(ii)* $G/(R \cdot \Xi)$ compact (product formula); *(iii)* B_χ is Ξ-invariant.

In the adelic case, $\Xi = \mathbb{Q}^\times$ satisfies (i)–(iii) with $\prod_v |x|_v = 1$ as product formula.

7.2 (FST-7) informational: Nyquist-Shannon

Definition 7.2 (information-theoretic self-duality)**.** (FST-7B): for all $f \in \mathcal{S}(G)$ with $\hat{f} \in \mathcal{S}(\hat{G})$,

$$\sum_{\xi \in \Xi} |f(\xi)|^2 = \sum_{\eta \in \hat{\Xi}} |\hat{f}(\eta)|^2.$$

Proposition 7.3. *(FST-7B) ⇔ (FST-7A)(i) under standard Pontryagin conditions on* (G, Ξ)*.*

Interpretation: Ξ is the Nyquist-optimal sampling of G.

Remark 7.4 (Slepian-Pollak prolates as a natural realisation)**.** The axiom (FST-7B) is compatible with dynamical realisations via prolate spheroidal wave functions $h_{n,\lambda}$ of Slepian-Pollak [14]: these simultaneously minimise out-of-interval and out-of-band energy on $\mathbb{R}$ and are therefore a Nyquist-Shannon-optimal basis; for $n \equiv 0 \pmod 4$ they are eigenfunctions of the Fourier transform, so they carry a concrete time-frequency self-duality. This is the conceptual entry point for the CCM programme (cf. Section 9). The *relation* between the prolate setup and (FST-7)(B) is a mathematically meaningful analogy rather than a proved instantiation; see Remark 9.3.

7.3 (FST-7) dynamical: arithmetic ergodicity

Definition 7.5 (arithmetic ergodicity)**.** (FST-7C): Ξ-action on the norm-1 kernel $G^{(1)}$ is uniquely ergodic (Haar measure).

Remark 7.6 (Conditional route from (FST-7C) to (FST-7A))**.** Unique or strict ergodicity of the Ξ-action is not by itself a general Poisson-summation theorem. The intended use of (FST-7C) is conditional: together with an invariant Haar measure, the relevant duality lattice, and the analytic hypotheses needed for the Poisson formula, it supplies the dynamical reading of (FST-7A).

In the adelic case, $\mathbb{Q}^\times$ acts uniquely ergodically on $\mathbb{A}^{(1)}$ by Strong Approximation [22].

7.4 Equivalence as structural conditions

Conditional Blueprint 7.7 (conditional structural schema for (FST-7) views)**.** *(FST-7A), (FST-7B), and (FST-7C) should be read as three compatible structural descriptions of the same supplied lattice/Poisson input, provided the relevant Pontryagin-duality, Haar-invariance, sampling, and Poisson-summation hypotheses have been independently verified. The schema is not an automatic theorem from the baseline FST axioms.*

Remark 7.8 (Scope of Schema 7.7)**.** The compatibility is at the level of structural input conditions, not at the level of technical entry points. In concrete cases the three views open very different technical doors: (A) opens Tate/Meyer adelic analysis; (B) opens Slepian-Pollack / prolate-operator analysis; (C) opens BC-thermodynamics / Cornelissen-Marcolli rigidity. The three entry points are *not* interchangeable as proof strategies.

7.5 (FST-7) is not derivable from baseline axioms

Structural observation. (FST-7) in any of its three forms is not derivable from (P1–P3), (SGE), (DS1–DS3), (RFEP) alone. It is an arithmetic property that must be supplied externally: from Tate's thesis in the adelic case, by analogous input in other cases. This is the core honest limitation of the axiom package.

8 The Bost-Connes/Meyer Bridge

8.1 The BC95 system

The Bost-Connes system [20] and its semigroup or Toeplitz-type variants [26, 27] motivate the QSM bridge used here. The Hecke C^*-algebra $\mathcal{H}_{\mathbb{Q}} = C^*(\mathbb{N}^\times \rtimes \hat{\mathbb{Z}}^\times)$ with time evolution $\alpha_t(\mu_n) = n^{it}\mu_n$ has:

- $\mathrm{spec}(H_{BC}) = \{\log n : n \in \mathbb{N}^\times\}$.
- $Z_{BC}(\beta) = \zeta(\beta)$ for $\beta > 1$.
- KMS states: unique for $\beta \le 1$, $\hat{\mathbb{Z}}^\times \cong \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$-parametrised for $\beta > 1$.
- Phase transition at $\beta = 1$.

8.2 Four bridge approaches, one recommended

8.3 Laplace-Mellin duality: the identification

$$L(\beta) := -\frac{d}{d\beta} \log Z_{BC}(\beta) = -\frac{\zeta'(\beta)}{\zeta(\beta)}. \tag{2}$$

Via the explicit formula, L is meromorphic with:

Approach	Plausibility	Short-term	Long-term
1. GNS + low-temperature limit	medium	medium	high
2. **Laplace-Mellin duality**	**high**	**high**	medium
3. Arithmetic Site intermediary	high	low	very high
4. Tropicalisation	low	low	medium

Table 2: Bridge approaches. Approach 2 is the recommended short-term target; it does *not* imply a proof of RH but provides a structural identification between the thermodynamic and distributional sides.

- Simple pole at $\beta = 1$ (Mangoldt pole), residue $+1$.
- Poles at $\beta = \rho$ (non-trivial zeros), residues $-m_\rho$.
- Poles at trivial zeros $\beta = -2k$.

Conjecture 8.1 (Meyer eigendistributions as BC residues)**.** *For Riemann zeta ($\chi = \chi_0$),*

$$v_{\gamma_\rho}(f) = \mathrm{Res}_{\beta=1/2+i\gamma_\rho} L(\beta) \cdot \mathcal{M}[f](\beta).$$

This is a conjectural identification, not a proved equivalence. Making it rigorous is exactly the content of the Wick-Mellin obstacles (Section 10); see there.

9 The CCM-Prolate Model: Closest Known Realisation of (FST-7)(B)

The Connes-Consani-Moscovici (CCM) programme [13, 12, 11] realises a rank-one perturbation of a scaling operator whose spectrum numerically converges to the non-trivial zeros of ζ. We summarise the construction and discuss its relation to (FST-7)(B); we then present the Route B blueprint for the CCM missing-step language, marking MS2/C2 as a conditional microcluster normal form tracked in the Zookeeper project and MS1 as the remaining item in this older route language. The internal study note [19] records the project-level mapping used in this section.

We emphasise at the outset: the Zookeeper material imported below is a conditional reduction template with open uniform constants, and the remaining material is a blueprint and/or an analogy.

9.1 The construction

Fix $\lambda > 1$ and set $L = 2\log\lambda$.

Scaling triple.
On $L^2([\lambda^{-1}, \lambda], d^*u)$ with $d^*u = du/u$, the scaling operator $D^{(\lambda)}_{\log} = -i\, d/d(\log u)$ with periodic boundary conditions has eigenfunctions $V_n(u) = L^{-1/2}\exp(2\pi i n \log(\lambda u)/L)$, $n \in \mathbb{Z}$.

Weil quadratic form.
From Weil's explicit formula the sesquilinear form $QW(f, g) = \Psi(f^* * g)$ with $\Psi = W_{0,2} - W_{\mathbb{R}} - \sum_p W_p$ is constructed. Restricted to $L^2([\lambda^{-1}, \lambda], d^*u)$, only primes $p \le \lambda^2$ contribute.

Test subspace E_N.
Span of V_n with $|n| \le N$; dimension $2N + 1$.

Rank-one perturbation.
Let ϵ_N be the smallest eigenvalue of the restriction QW^N_λ, and ξ the corresponding eigenvector (assumed simple and even under $u \mapsto u^{-1}$). Let $\delta_N \in E_N$ represent the Dirichlet kernel evaluation at the boundary, normalised by $\delta_N(\xi) = 1$. Define

$$D^{(\lambda,N)}_{\log} := D^{(\lambda)}_{\log} - |D^{(\lambda)}_{\log}\xi\rangle\langle\delta_N|.$$

Theorem 9.1 (CCM 2025, Theorem 1.1). *(i) $D^{(\lambda,N)}_{\log}$ is self-adjoint on $E'_N \oplus E^{\perp}_N$ with $E'_N = E_N/\mathbb{C}\xi$. (ii)* $\det_{\mathrm{reg}}(D^{(\lambda,N)}_{\log} - z) = -i\lambda^{-iz}\widehat{\widehat{\xi}}(z)$. *(iii) $\widehat{\widehat{\xi}}(z)$ is entire, all zeros are on the real axis, and coincide with* $\operatorname{spec}(D^{(\lambda,N)}_{\log})$.

Remark 9.2 (Numerical evidence). For $\lambda = \sqrt{14}$, $N = 120$, using only primes $p \leq 13$, the first fifty non-trivial zeros of $\zeta(1/2 + is)$ are recovered with errors ranging from 10^{-60} to 10^{-6} (cf. [11, Table 1]). The statistical improbability of such coincidence is estimated by CCM at 10^{-1235}. Numerics of this strength are *compelling evidence*; they are not a theorem reducing RH.

9.2 The QW-PW duality

CCM identify a structural duality between the Weil quadratic form QW_λ and the prolate wave operator

$$PW_\lambda := -\partial_x((\lambda^2 - x^2)\partial_x) + (2\pi\lambda x)^2,$$

whose eigenfunctions $h_{n,\lambda}$ are the prolate spheroidal wave functions of Slepian-Pollak [14].

	Weil QW_λ	**Prolate** PW_λ
Origin	Number theory (Weil)	Information theory (Shannon)
Regime	Infrared (low eigenvalues)	Ultraviolet (high eigenvalues)
Zeta imprint	Low zeros of ζ	UV behaviour of zeros
Optimality	Explicit-formula positivity	Time-frequency optimal
Reference	[11]	[12]

9.3 CCM as the closest known model for (FST-7)(B)

Recall (FST-7)(B) from Section 7: there exists a *Nyquist-Shannon-optimal* discrete subset $\Xi \leq G$ on which a Plancherel-type identity holds. The prolate spheroidal wave functions of Slepian-Pollak *are* the canonical Nyquist-Shannon-optimal basis in the time-frequency plane.

Remark 9.3 (CCM as the closest known model for (FST-7)(B)). The CCM construction $(D^{(\lambda,N)}_{\log}, QW_\lambda, PW_\lambda)$ is a *closely related concrete model* of (FST-7)(B) in the sense that (a) it is built around the canonical Slepian-Pollak Nyquist-Shannon-optimal basis in the time-frequency plane, and (b) the scaling action of $\mathbb{R}^*_+$ on $L^2([\lambda^{-1}, \lambda], d^*u)$ provides the locally compact abelian setting required by the (FST-7)(B) formulation. This relation is a mathematically meaningful analogy that aligns the abstract optimality statement with a concrete operator-theoretic setting; it is *not* a proved theorem-level equivalence. A rigorous statement would require constructing an explicit Ξ from prolate data, proving a Plancherel-type identity in the sense of Definition 7.2, and relating this to the rank-one perturbation condition on δ_N. None of these are established here.

Conjecture 9.4 ($QW \leftrightarrow PW$ as (FST-7)(A)$\leftrightarrow$(FST-7)(B) equivalence operator, structural form). *The CCM duality $QW_\lambda \leftrightarrow PW_\lambda$ between infrared and ultraviolet regimes is conjectured to be a dynamical representation of the abstract equivalence (FST-7)(A) $\Leftrightarrow$ (FST-7)(B) from Schema 7.7. Concretely: there should exist a transformation sending QW_λ-eigenfunctions to PW_λ-eigenfunctions in the $\lambda \to \infty$ limit that realises the Pontryagin self-duality of $\Xi = \mathbb{Q}$ inside $\mathbb{A}_{\mathbb{Q}}$ at the level of spectral measures. The present paper provides a candidate for this transformation (Section 9.5) but does not prove the intertwining property.*

9.4 The CCM missing steps

Two residual technical gaps were identified in the CCM programme:

(MS1)

Simple-even at the smallest eigenvalue of QW_λ. Known classically for the prolate operator PW_λ [14]; numerically confirmed for QW_λ; not yet proved. Remains open.

(MS2)

Approximation quality. The Poisson quasimode k_λ must approximate the true QW_λ-eigenfunction ξ_λ well enough to transfer spectral convergence. **Reduced in the Zookeeper paper to conditional uniform constants** via the Three-Lemma Microcluster architecture: scalar secular cancellation, projected Poisson quasimode control, and a coercive complement give the finite-window estimate $\|(I - P_{V_\lambda})k_\lambda\| \le r_\lambda / g_*$. The limiting statement still requires the uniform inputs $s_\lambda \to 0$, $p_\lambda \to 0$, and $g_* \ge g_0 > 0$.

(MS1) remains an open problem in this older route language. (MS2) is no longer a formless obstruction in the CORE, but it is also not a closed theorem-level step: the Zookeeper paper records the conditional microcluster reduction of the approximation step in the CCM-Fourier model, while uniform constants and the surrounding CCM transfer remain explicit gates.

9.5 FST-transfer: a conditional reduction blueprint

We now make Conjecture 9.4 concrete enough to state a *conditional reduction blueprint* for the two CCM gaps. This subsection is *not* a theorem and does *not* prove a reduction; it is a scheme in which each step is open and clearly identified.

9.5.1 Two candidate isometries

Let $H_{PW} = L^2([-\lambda, \lambda], dx)$ with reflection $\rho(g)(x) = g(-x)$ and $H_{QW} = L^2([\lambda^{-1}, \lambda], d^*u)$ with inversion $\iota(f)(u) = f(u^{-1})$. The logarithm-plus-rescaling map

$$\Phi_\lambda := \log^* \circ \sigma_\lambda : L^2([-\lambda, \lambda], dx) \xrightarrow{\sigma_\lambda} L^2([-\log\lambda, \log\lambda], dx) \xrightarrow{\log^*} L^2([\lambda^{-1}, \lambda], d^*u)$$

is a *kinematic* isometry intertwining ρ and ι — a direct computation.

Building on the explicit CCM construction we *propose* as candidate dynamical map

$$\Psi_\lambda : H_{PW} \longrightarrow H_{QW}, \qquad g \mapsto (\mathcal{E}(g))\big|_{[\lambda^{-1},\lambda]}, \qquad \mathcal{E}(g)(u) := u^{1/2} \sum_{n\ge1} g(nu),$$

i.e. restriction to $[\lambda^{-1}, \lambda]$ of the Epstein-Mellin convolution. Whether Ψ_λ is bounded as an operator from H_{PW} to H_{QW}, whether it has dense range, and what its precise adjoint and intertwiner properties are, are themselves non-trivial questions. We collect these as the first open item:

Milestone 9.5 (Domain and form theory for Ψ_λ)**.** *Establish that Ψ_λ is a well-defined, densely defined operator $H_{PW} \to H_{QW}$ and determine its form domain, boundedness class, and intertwiner structure with respect to the involutions (ρ, ι). Prove or refute that $\Psi_\lambda^* \Psi_\lambda$ has compact resolvent.*

9.5.2 Conditional reduction blueprint

Conditional Blueprint 9.6 (MS1 $\Leftarrow$ (MS2 $\wedge$ bounded intertwining))**.** *Suppose that Milestone 9.5 has been resolved such that Ψ_λ is a bounded intertwiner on appropriate form domains. Suppose further there exist constants $\mu_\lambda > 0$ and $\varepsilon_\lambda \to 0$ as $\lambda \to \infty$ such that*

$$\|QW_\lambda \circ \Psi_\lambda - \mu_\lambda \Psi_\lambda \circ PW_\lambda\| \;\le\; \varepsilon_\lambda \tag{3}$$

as operators on the appropriate dense domain (cf. (MS2)). Then the following is a conditional conclusion, not a theorem: by Slepian-Pollak [14], the smallest eigenvalue $\mu_{0,\lambda}$ of PW_λ is simple with even eigenfunction $h_{0,\lambda}$; under (3) *and the intertwining hypothesis, a Kato-type perturbation argument [17] would give a simple QW_λ-eigenvalue ϵ_λ with eigenvector ξ_λ satisfying $\|\xi_\lambda - c \cdot \Psi_\lambda h_{0,\lambda}\| \le C\varepsilon_\lambda$, and the preservation of the ρ/ι grading would give the even property. In that scenario (MS1) would follow whenever (MS2) holds.*

Remark 9.7 (Status of Blueprint 9.6)**.** The conditional conclusion in Blueprint 9.6 rests on three open items: (1) Ψ_λ is actually a bounded intertwiner (Milestone 9.5); (2) the operator-norm bound (3) holds with the claimed decay; (3) the Kato perturbation argument applies uniformly on the relevant form domains. Each item is non-trivial. In particular the standard reading of Kato's theorem requires operator/form convergence in a norm or form-resolvent sense that is *not* automatic from (3). The blueprint is best read as an articulation of what a proof of MS1 via MS2 would have to establish, not as a reduction theorem.

Remark 9.8 (Role of FST in the blueprint)**.** Even in its conditional form, Blueprint 9.6 illustrates the way FST is intended to contribute: the axiom equivalence (FST-7)(A)$\leftrightarrow$(FST-7)(B) structurally motivates the search for an intertwiner Ψ_λ between the two concrete operators on the CCM side; the CCM construction itself supplies the candidate explicit form of Ψ_λ. What is missing is the analytic content (boundedness, operator-norm control, application of perturbation theory).

9.6 Milestone agenda for Route B

Replacing the heuristic Conjecture ("Weyl law for QW_λ") of v0.6 with a structured milestone agenda, Route B decomposes into four concrete, sequential milestones in the sense of the audit recommendation:

Milestone 9.9 (Domain / form theory for Ψ_λ)**.** *(= Milestone 9.5.) Establish that Ψ_λ is a well-defined, boundedly invertible intertwiner on appropriate form domains. First genuine rigorous step of Route B.*

Milestone 9.10 (Low-eigenvalue comparison via min-max)**.** *Prove a min-max comparison between the first finitely many eigenvalues of QW_λ and PW_λ,* without *requiring a full Weyl asymptotic. This is the operator-theoretic minimum needed to attack (MS1) uniformly.*

Milestone 9.11 (Finite-dimensional CCM truncation)**.** *Prove a finite-dimensional CCM truncation theorem inside E_N: control the restriction of QW_λ to E_N and its comparison to the prolate restriction, using standard finite-dim perturbation (Kato's theorem applies cleanly there). This is where the CCM numerics actually live.*

Milestone 9.12 (Asymptotic regime)**.** *Only after Milestones 9.9–9.11 are in place: investigate the asymptotic behaviour of $N_{QW_\lambda}(\mu)$ as $\mu \to \infty$ and $\lambda \to \infty$. This is the analogue of what v0.6 stated speculatively as a "Weyl law for the Weil form" and is here explicitly marked as the* last *milestone in the sequence, not the first.*

Remark 9.13 (Why this ordering)**.** Milestones 9.9–9.11 are finite-dimensional or form-theoretic in character and are genuinely tractable with existing operator-theory machinery. Milestone 9.12 involves genuine global asymptotics of a non-standard quadratic form and is much less well understood. The v0.6 draft treated Milestone 9.12 as if it were a near-theorem; the present revision explicitly reorders the agenda so that the tractable work comes first and the deep asymptotic item last. See the external audit notes for the motivation of this reordering.

9.7 The Three-Lemma Endgame for Route B

We now present a conditional microcluster estimate for (MS2) via a finite-dimensional spectral analysis of the CCM rank-one operator. The argument requires three quantitative inputs and produces a quasimode-to-projector bound as a corollary; the limiting MS2 claim requires separate uniform control.

9.7.1 Setup and notation

Fix $\lambda > 0$. Let $L = 2\log\lambda$, $N = \max(40, \lceil 12L \rceil)$, and let $E_N \subset L^2[0, L]$ be the Fourier–Galerkin subspace spanned by $\{e_n(x) = \sqrt{2/L}\sin(n\pi x/L)\}_{n=1}^{N}$. Within E_N the CCM operator takes the form

$$A^{(N)} \;=\; H^{(N)} + \tilde{C}\,|\tilde{u}\rangle\langle\tilde{u}|, \tag{4}$$

where $H^{(N)}$ is the Galerkin restriction of the kinetic-plus-potential part and $\tilde{u} \in E_N$ encodes the Poisson-kernel interaction. The Poisson quasimode $k = k_\lambda^{(N)} \in E_N$ is the normalised Galerkin projection of the Poisson kernel $K(e^{x/\lambda})$. We write u_0 for the smallest eigenvalue of $A^{(N)}$ and $\{(u_j, q_j)\}_{j=0}^{N-1}$ for the full eigensystem.

Hypothesis 9.14 (Galerkin faithfulness)**.** *For each λ, the finite-dimensional Galerkin model $A^{(N(\lambda))}$ with $N = \max(40, \lceil 12L \rceil)$ faithfully represents the spectral cluster structure of the CCM operator in the following sense: the eigenvalue ordering, the rank-one secular equation, and the Cauchy interlacing with the background operator $H^{(N)}$ hold with errors controlled by the Galerkin truncation error $\varepsilon_{\mathrm{Gal}} \sim e^{-c\cdot N/L}$.*

Remark 9.15. Hypothesis 9.14 is numerically verified to $\varepsilon_{\mathrm{Gal}} < 10^{-12}$ for the tested values

$$\lambda \in \{3, 5, 7, 10, 15, 20, 30, 40, 50, 75, 100\}.$$

A rigorous proof would require a Galerkin convergence theorem for the specific rank-one operator (4), which is the subject of ongoing work.

9.7.2 Diagnostic mass-based effective gap

Definition 9.16 (Diagnostic effective cluster gap)**.** For $\varepsilon > 0$, let $J_\varepsilon = \min\{J : \sum_{j=0}^{J} |\langle q_j, k\rangle|^2 \geq 1 - \varepsilon\}$ be the spectral mass threshold. The *effective gap* is

$$g_{\mathrm{eff}}(\varepsilon) \;:=\; u_{J_\varepsilon+1} - u_0.$$

The resonant subspace is $V_\lambda := \mathrm{span}\{q_j : j \leq J_\varepsilon\}$.

This definition is a posterior diagnostic because the subspace is selected by the same quasimode mass that it is later used to assess. It replaces fixed-tolerance cluster identification (which produced erratic gaps $\sim 10^{-6}$–10^{-4}) with a spectral-mass criterion that yields $g_{\mathrm{eff}}(\varepsilon) \approx 5$–$7$ uniformly for all tested λ and all $\varepsilon \leq 10^{-4}$. The key point: the quasimode mass $M_k(J) = \sum_{j \leq J} |\langle q_j, k\rangle|^2$ saturates within the first 1–3 eigenvalues, exhibiting an order-one separation in the tested windows. It is not yet an independent outer-gap certificate.

9.7.3 The three lemmas

Lemma 9.17 (Scalar Secular Cancellation (C2ca.1, conditional))**.** *Let $\mu = \langle\tilde{u}, (A - u_0)k\rangle$ be the scalar component of the residual along $\tilde{u}$. If a Galerkin truncation bound s_λ is available, then*

$$|\mu| \;\leq\; s_\lambda.$$

Under Hypothesis 9.14, the rank-one secular equation $1 + \tilde{C} \cdot m_H(w) = 0$ produces two terms of magnitude $\sim 10^{-2}$ that cancel to $\sim 3 \times 10^{-7}$ (a factor > 200 cancellation) in the tested Galerkin regime. The theorem-level claim $s_\lambda \to 0$ is an additional uniform truncation input.

Proof sketch. The secular identity (from the rank-one resolvent formula) gives

$$\frac{\mu}{\alpha} = (\bar{w} - u_0) + \frac{\langle f_\lambda, k_\perp\rangle}{\alpha},$$

where $\alpha = \langle\tilde{u}, k\rangle$, $\bar{w}$ is the secular root, and f_λ encodes the H-spectral distribution of $\tilde{u}$. Both terms are $O(10^{-2})$ but carry opposite signs due to the interlacing structure $t_j \leq u_j \leq t_{j+1}$ (where t_j are H-eigenvalues). The cancellation is controlled by the Galerkin truncation error. □

Lemma 9.18 (Projected Poisson Quasimode (C2ca.2, conditional))**.** *Let $h = P_0(A - u_0)k$ be the $\tilde{u}$-orthogonal component of the residual, where $P_0 = I - |\tilde{u}\rangle\langle\tilde{u}|$. If the projected bulk-leakage and boundary terms are controlled by a quantity p_λ, then*

$$\|h\| \;\leq\; p_\lambda,$$

where $p_\lambda \approx 2$–3×10^{-6} is a stable numerical value at fixed tested N/L. The limiting claim $p_\lambda \to 0$ remains a separate uniform estimate.

Proof sketch. The boundary-defect decomposition (from the Poisson transfer theory) gives $h = \frac{1}{n_{L,N}} P_0 \Pi_{L,N}(E_L^{\mathrm{bulk}} + B_L)$, where E_L^{bulk} is $> 99.999\%$ in the $\tilde{u}$-direction (its P_0-leakage is $< 5 \times 10^{-7}$), and B_L is a boundary-localised term controlled by the exponential decay of the Poisson kernel at $x = L$. □

Lemma 9.19 (Coercive Complement (C2ca.5, conditional))**.** *Under Hypothesis 9.14 and an additional uniform outer-gap certificate, there exists $g_* > 0$ such that for all $v \in V_\lambda^\perp$ with $\|v\| = 1$:*

$$\langle (A^{(N)} - u_0)\, v,\, v\rangle \;\geq\; g_*.$$

Numerically, the diagnostic mass-defined gap satisfies $g_* \approx 5$–7 for all tested λ. This posterior gap motivates, but does not replace, the additional independent outer-gap certificate required above.

Proof sketch. For a fixed preselected spectral window, the min–max principle gives the finite identity

$$\inf_{\substack{v \perp V_\lambda \\ \|v\|=1}} \langle (A - u_0)v, v\rangle = u_{J+1} - u_0.$$

Rank-one Cauchy interlacing can transfer a gap from the background operator $H^{(N)}$, but it does not by itself prove an order-one, uniform-in-λ lower bound. In particular, the crude potential-free Laplacian floor $\pi^2 L^{-2}((J+1)^2 - 1)$ is only an $O(L^{-2})$ baseline and is not ≥ 5 for the full tested L-range. The values $g_* \approx 5$–7 are therefore numerical diagnostics of the full mass-defined cluster gap, pending a separate fixed-waterline or outer-gap proof. □

9.7.4 Mass concentration as corollary

Theorem 9.20 (Mass Concentration — Conditional Microcluster Estimate)**.** *Under Hypothesis 9.14, and with the three conditional inputs above:*

$$\|(I - P_{V_\lambda})\, k_\lambda\| \;\leq\; \frac{r_\lambda}{g_*} \;\leq\; \frac{s_\lambda + p_\lambda}{g_*} \;\approx\; 6 \times 10^{-7},$$

where $r_\lambda = s_\lambda + p_\lambda$ is the total residual bound from Lemmas 9.17 and 9.18, and g_* is the coercive gap from Lemma 9.19.

Proof. This is the standard quasimode-to-projector argument. Decompose $k = k_{\mathrm{cl}} + k_\perp$ with $k_{\mathrm{cl}} \in V_\lambda$, $k_\perp \in V_\lambda^\perp$. Since V_λ is A-invariant:

$$\begin{aligned} g_* \|k_\perp\|^2 &\leq \langle k_\perp, (A - u_0)k_\perp\rangle = \langle k_\perp, (A - u_0)k\rangle \\ &\leq \|k_\perp\| \cdot \|(A - u_0)k\| = \|k_\perp\| \cdot R_0. \end{aligned}$$

If $\|k_\perp\| > 0$, divide by $\|k_\perp\|$: $\|k_\perp\| \leq R_0/g_*$. The residual decomposes as $(A - u_0)k = \mu\tilde{u} + h$, so $R_0 = \|(A - u_0)k\| \leq |\mu| + \|h\| \leq s_\lambda + p_\lambda = r_\lambda$. □

Remark 9.21 (Convergence to zero)**.** At fixed $N/L = 12$, the bound $r_\lambda/g_* \approx 6 \times 10^{-7}$ is already small but does not tend to zero. For $\|(I - P_{V_\lambda})k\| \to 0$ as $\lambda \to \infty$ (as required by the CCM programme), one would choose $N = N(\lambda)$ with $N/L \to \infty$ (e.g., $N = L^{1+\delta}$). Then $r_\lambda^{(N)} \sim e^{-c \cdot N/L} \to 0$; the conclusion $\|k_\perp\| \to 0$ follows only if an independent uniform lower bound $g_* \geq g_0 > 0$ is proved in that same scaling regime.

Remark 9.22 (Numerical diagnosis)**.** In the tested finite windows, the bound in Theorem 9.20 is 75–84% tight: the actual $\|k_\perp\|$ values are within a factor 0.75–0.84 of the upper bound R_0/g_*, confirming near-optimality of the quasimode-to-projector estimate. Full spectral mass diagnostics (eigenvalue spectra, overlap coefficients, cumulative mass curves) are computed for $\lambda = 3, \dots, 100$ and available in the accompanying repository.

9.7.5 Structural summary

The proof architecture is:

$$\underbrace{\text{C2ca.1 (Secular)}}_{\text{Lemma 9.17}} + \underbrace{\text{C2ca.2 (Projected)}}_{\text{Lemma 9.18}} \longrightarrow R_0 \le r_\lambda \xrightarrow{\text{Lemma 9.19 } (g_*)} \|k_\perp\| \le r_\lambda/g_*.$$

Three conditional quantitative inputs and one standard inequality. The previous four-lemma structure (where "mass concentration" was listed as an independent fourth lemma) is superseded: the fourth lemma is a corollary, not an input.

9.8 RFEP transfer package for physical supplements

The Three-Lemma Endgame is not only a Route B device. It also supplies a reusable RFEP pattern for the physical side of the FST programme. The physical supplements should not import the Riemann proof attempt verbatim; rather, they should import the operator grammar:

$$A_X^{(N)} = H_X^{(N)} + B_X^{(N)}, \qquad \|(I - P_{V_X})k_X\| \le \frac{r_X}{g_X}.$$

Here X denotes a physical branch such as Yang–Mills, Navier–Stokes, NS-LDI, turbulence, or CRM; $B_X^{(N)}$ is the rank-one or finite-rank defect; k_X is the RFEP-selected candidate; r_X is its residual; and g_X is the coercive gap on the orthogonal complement. The working note [3] records this as the RFEP–Zookeeper normal form.

The v1.10 candidate guardrail refines the last quotient before it is used as evidence. In constrained or projected problems the ledger separates the raw total residual from the evidence-bearing tangent component:

$$r_X^{\text{tot}} = r_X^{\text{tan}} + r_X^{\text{normal,paid}} + r_X^{\text{escape}}.$$

Only $r_X^{\text{tan}}/g_X^{\text{tan}}$ can support a transfer reading, and only after the normal component is paid by explicit constraints and the target/projection/gap provenance is independent. A small r_X^{tot}/g_X value by itself remains a diagnostic number, not certificate evidence.

Proposition 9.23 (RFEP transfer schema, conditional)**.** *Let X be a physical FST supplement. Suppose that:*

1. *the RFEP functional for X admits a closed quadratic or operator representative A_X on a form domain V_X;*
2. *there is a Galerkin exhaustion $A_X^{(N)}$ faithful to the relevant low spectral cluster;*
3. *$A_X^{(N)}$ decomposes as a coercive background $H_X^{(N)}$ plus a rank-one or finite-rank defect $B_X^{(N)}$;*
4. *the RFEP candidate k_X has residual $\|(A_X^{(N)} - u_0)k_X\| \le r_X = s_X + p_X$ with a scalar secular part and a projected quasimode part; in constrained branches the ledger must further split total residual, projected/tangential residual, paid normal residual, and unpaid escape;*
5. *on $V_X^\perp$ there is a coercive complement gap $g_X > 0$.*

Then

$$\|(I - P_{V_X})k_X\| \le r_X/g_X.$$

If, in addition, the target cluster is independently predefined and simple or RFEP-selective, the stable physical state is unique within the corresponding Galerkin sector.

Proof. The proof is identical to Theorem 9.20: decompose k_X into cluster and complement components, use coercivity on the complement, and bound the complement component by the residual. No physics-specific conclusion follows unless the five hypotheses are verified in the given domain. □

Remark 9.24 (Independent certificate requirement)**.** The projector P_{V_X} must not be certified only by the same diagonalisation, fit, or posterior mass concentration that defines the candidate k_X. A domain transfer is certificate-level only if the target subspace, outer waterline, or complement gap is fixed by an independent source: symmetry, a background operator, a pre-registered spectral window, a finite certificate bridge, or a Galerkin/continuum estimate not fitted to the same residual. Otherwise the result is a posterior self-consistency diagnostic, not a domain closure.

Remark 9.25 (Certificate ledger before residual evidence)**.** A residual-over-gap estimate is only a leakage diagnostic until the certificate ledger is paid. For a domain transfer the target subspace must be fixed before inspecting Ritz mass; the source axis must be independent of the claimed target; the outer gap must come from an independent source; the Galerkin sequence must carry an exactness or no-pollution certificate; and the projected residual must be tested against matched negative controls. In constrained problems the active constraints, projected gradient or projected residual, tangent Hessian or resolvent status, paid normal residual, unpaid escape directions, and sloppy/null directions must be recorded separately. Only after these fields are visible may a small projected residual over a tangent-gap lower bound be used as evidence for RFEP selection. A small raw total residual, or a large total residual whose normal part has not been paid by constraints, remains diagnostic only.

Ledger block	Required visible fields and evidence gate
Identifier and target	`target_id`, `target_predefined`, `target_source`, `source_axis_independent`, and `projection_predefined`. A posterior target or projector keeps the row diagnostic.
Gap certificate	`outer_gap_source`, `gap_lower_bound_delta`, `truncation_error_bound`, `contour_resolvent_bound`, `riesz_projector_stability`, and `spectral_exactness_status`. The admissible source is independent, for example `resolvent_contour`.
Tangential residual	`total_residual_over_gap`, `projected_residual_over_tangent_gap`, `tangent_gap_lower_bound`, and `projected_residual_source`. Only this projected quotient can support a transfer reading.
Normal and escape account	`normal_residual`, `normal_paid_by_constraint`, `active_constraint_set`, `unpaid_escape_status`, and `sloppy_null_direction_class`. Any unpaid escape blocks promotion.
Scale and controls	`detector_degree`, `target_degree`, `capacity_ratio`, `advice_source`, and `matched_control_family`. The negative controls must be matched to the same target and projection rules.
Decision	`decision`, `decision_basis`, `blocking_field`, and `promoted_to_certificate`. Empty or unknown gate fields default to diagnostic status.

Table 3: RFEP v1.10 tangential-provenance ledger for physical-domain rows. The table turns the residual claim into explicit data fields: target provenance, gap certificate, tangential residual, paid normal component, matched controls, and decision. The mechanical positive-control rows in the project ledger are controls, not completed physical-domain certificates. As of this revision, no physical branch row is promoted beyond diagnostic status by this table.

Table 3 is deliberately read as a data-entry surface. It prevents a physical supplement from reporting a favourable raw residual without also exposing the target provenance, the independent gap certificate, the tangential projection, the paid normal component, and the matched controls that make the number interpretable.

The control family behind Table 3 contains two mechanical `candidate_pass_not_theorem` rows (`tangent_projected_positive_control` and `large_normal_paid_small_tangent`) and six blocking rows: `raw_residual_looks_good_but_unprojected`, `unpaid_normal_leak`, `posterior_tangent_target`, `spectral_pollution_unknown`, `matched_control_missing`, and `sloppy_null_in_tangent`. Their role is to test the ledger bookkeeping itself; they are not physical-domain successes.

Remark 9.26 (What RFEP gains)**.** RFEP gains a concrete proof checklist. A physical paper can now ask: what is the operator or form A_X; what is the defect B_X; what is the Galerkin faithfulness statement; where is the complement gap; and how does the domain limit preserve the RFEP selection? This is stronger than a narrative analogy but weaker than a domain theorem.

Remark 9.27 (Domain readings)**.** For Yang–Mills the target reading is mass gap or Gibbs uniqueness as a coercive complement. For Navier–Stokes and NS-LDI it is dissipative selection and log-distance defect control as a stable cluster statement. For turbulence it is the K41 spectrum as a mass-concentrated RFEP minimiser. For CRM the main proof language remains strict convexity/UCU, but the transfer package supplies operator language for perturbative or boundary-coupled variants. The Prime-Hub programme [4] is the natural intermediate bridge: frontier-prime insertions and bouquet truncations are already close to finite-rank boundary defects.

9.9 The Sonin-space bridge (context from CCM 2024)

The operator-norm estimate (3) sits inside the analytic context provided by the Connes-Consani-Moscovici *semilocal trace formula* [15]:

Theorem 9.28 (CCM 2024, semilocal decomposition)**.** *In the archimedean case the prolate operator admits the decomposition*

$$PW_\lambda \;=\; D_{\log}^{(\lambda)2} \;+\; \Gamma,$$

where Γ *is a grading operator on orthogonal polynomials. Correspondingly its spectrum splits as*

- *a* positive part *realising low-lying zeros of* $\zeta(1/2+is)$ *(infrared / Weil regime),*
- *a* negative part *on the Sonin space realising the UV asymptotic distribution of zeros (ultraviolet / Connes-Moscovici 2022 regime).*

Under this decomposition the Weil form QW_λ coincides with the restriction of PW_λ to the positive spectral subspace, up to the rank-one correction encoded by the trace-formula error $W_\infty - W_0$. This identifies the operator difference in (3) with the Connes-Consani arithmetic-archimedean defect [16]. Whether this defect admits the kind of operator-norm bound that Blueprint 9.6 requires is an open question.

10 The Wick-Mellin Rotation and Its Three Obstacles (Route A)

10.1 The rotation is not standard Wick rotation

Standard Wick rotation connects $\operatorname{Re}\beta > 0$ (thermal) with $\operatorname{Im}\beta \in \mathbb{R}$ (unitary). What Route A wants is different: $\operatorname{Re}\beta > 1$ (BC Gibbs) to $\operatorname{Re}\beta = 1/2$ (Meyer critical line), i.e. *half-plane continuation through the Mangoldt pole at* $\beta = 1$.

The Gibbs operator $e^{-\beta H_{BC}}$ is:

- Trace-class for $\operatorname{Re}\beta > 1$.
- Bounded but not trace-class for $0 < \operatorname{Re}\beta \leq 1$.

On the critical line, the classical Gibbs state does not exist; only the meromorphically continued $\zeta(\beta)$ persists. Three concrete obstacles structure the open technical work.

10.2 Obstacle (O1): The distribution class

We seek a precise Fréchet-space $\mathcal{V}_\chi$ whose dual contains Meyer's eigendistributions.

Definition 10.1 (the Meyer space)**.** $\mathcal{V}_\chi := \mathcal{S}_0(\mathbb{A}^\times)^\chi/\mathcal{K}^\chi_{\mathbb{Q}^\times}$, where $\mathcal{S}_0 = \{f : Z(f,s) \text{ entire}\}$ and $\mathcal{K}_{\mathbb{Q}^\times}$ the $\mathbb{Q}^\times$-coinvariant closure.

Seminorms: archimedean weighted Schwartz norms $\|f_\infty\|_{n,m} = \sup|x^m\partial^n f_\infty(x)|$; p-adic norms $\|f_p\|^p_N$ on compact subsets of $\mathbb{Q}_p^\times$. $\mathcal{V}_\chi$ is Fréchet in the quotient topology.

$\mathbb{Q}^\times$-invariance of Mellin evaluation. For $f \in \mathcal{S}_0$, $q \in \mathbb{Q}^\times$:

$$v_\gamma^{(pre)}(T_q f) = |q|^{1/2+i\gamma} \cdot v_\gamma^{(pre)}(f) = v_\gamma^{(pre)}(f)$$

by the product formula $|q| = 1$. This is exactly where (FST-7A)(ii) enters.

Status of (O1). The distribution class is closely tied to recognised adelic analysis and Meyer 2005; the Fréchet-quotient and Mellin-evaluation picture is mathematically recognisable. The real analytic content of (O1) is the density/completeness of Mellin evaluations in the dual of $\mathcal{V}_\chi$ — essentially a completeness theorem that must be imported from, or proved in the style of, Meyer 2005, Sections 4–5. This is comparatively solid relative to (O2), (O3), but the abstract-generalisation burden is non-trivial.

10.3 Obstacle (O2): The Fourier duality

We seek an explicit operator-algebraic Fourier transform

$$\mathcal{F} : \mathcal{H}^{\mathrm{Sch}}_{\mathbb{Q}} \to \mathcal{V}_{\chi_0}$$

between the Bost-Connes Hecke algebra (Schwartz subalgebra) and Meyer's Fréchet space.

Four bridge statements (targets, not theorems):

- (F1) Element correspondence: $\mu_n \leftrightarrow \mathbf{1}_n$.
- (F2) State correspondence: KMS state $\phi_{\beta,\sigma} \leftrightarrow$ Galois-twisted Tate functional.
- (F3) Time evolution: $\alpha_t^{BC} \leftrightarrow$ Mellin shift via analytic continuation $t = i(\beta - 1/2)$.
- (F4) Spectrum matching: $\{\log n\} \leftrightarrow$ continuum spectrum with singular points at $\{\gamma_\rho\}$.

Candidate construction via Galois averaging:

$$\mathcal{F}(a)([f]) := \int_{\hat{\mathbb{Z}}^\times} \phi_{\infty,\sigma}(a) \cdot \sigma(f)\, d\sigma.$$

Status of (O2). A concrete target map is described, but the note-level statements for (F1)–(F4) lack proofs of existence, continuity, injectivity, and surjectivity. The limiting KMS states and their continuity are not constructed. The proposed Fourier map remains heuristic exactly at the point where the critical-line continuation should enter. (O2) should therefore be treated as a standalone research problem in operator algebras rather than as a quick bridge lemma.

10.4 Obstacle (O3): Convergence control

The Mangoldt pole at $\beta = 1$ is an obstruction to direct operator continuation. A renormalisation approach suggests itself, but — as the audit correctly emphasises — neither the trace-class properties nor the analytic continuation of the renormalised operator are currently justified by rigorous arguments.

Definition 10.2 (renormalised Gibbs operator, tentative)**.**

$$G_\beta^{\mathrm{ren}} := e^{-\beta H_{BC}} - \frac{\Pi_0}{\beta - 1},$$

where Π_0 is the canonical projector onto the "$n = 1$" identity sector of the Hecke algebra (carrying the Mangoldt pole).

Conjecture 10.3 (Trace-class continuation through the pole, **open**))**.** G_β^{ren} *is trace-class for all* $\mathrm{Re}\,\beta > 0$ *(including the critical line) with* $\mathrm{Tr}(G_\beta^{\mathrm{ren}}) = Z_{\mathrm{ren}}(\beta) = \zeta(\beta) - (\beta - 1)^{-1}$.

Remark 10.4 (Retraction of the v0.6 proposition)**.** In v0.6 Conjecture 10.3 was stated as a proposition. This was incorrect: subtracting a pole at the level of traces does not automatically produce a trace-class operator family. An operator-theoretic proof requires, at minimum, explicit analysis of the rank-one subtraction on the Hilbert space of the BC representation, control of operator-norm limits as $\mathrm{Re}\,\beta \searrow 1$, and a rigorous identification of the continued family on $\mathrm{Re}\,\beta \leq 1$. None of this is supplied by the scalar zeta-regularisation. We retract the v0.6 claim and mark Conjecture 10.3 explicitly as open.

FST interpretation. The Mangoldt-pole subtraction is suggestive of an RG operation separating bulk free energy from the RFEP-stable part. The remaining pole structure $\{\beta = \rho\}$ would, conjecturally, give Meyer's eigendistributions as the RG-critical modes of a renormalised system. This remains a structural picture, not a theorem.

10.5 Status of the three obstacles

We do *not* claim that (O1)–(O3) reduce to standard techniques. The honest assessment is:

- (O1) is comparatively solid — close in style to Meyer 2005 and classical adelic analysis; the abstract generalisation is non-trivial but feasible.
- (O2) is open but structured: a concrete target map is specified, but the analytic ingredients (continuity, limiting KMS states, injectivity, surjectivity) are not in place.
- (O3) is deep and highly speculative in its current form. The trace-class continuation through the pole has been claimed too strongly in earlier versions; see Remark 10.4.

The sentence "all three obstacles reduce to standard techniques", present in v0.6, is not accurate and is withdrawn.

11 Non-abelian Extension: The Selberg Compatibility Check

11.1 Why a compatibility check matters

The baseline FST axioms (Section 3) require G abelian. Selberg's trace-formula proof of RH for Z_Γ [24] is a *classical* non-abelian case. If a natural non-abelian extension of the FST axioms is *compatible* with Selberg's setting — that is, if the axioms can be translated to the non-abelian setting without contradiction and the Selberg structures fit the translated axioms — this is evidence that the axioms do not have an obviously abelian-only structural bias.

We emphasise: a compatibility check is not a proof. The non-abelian FST fixed-point theorem (Conjecture 11.1 below) is a conjecture, not a theorem recovering Selberg from the new axioms.

11.2 Selberg recap

For $\Gamma \subset \mathrm{PSL}_2(\mathbb{R})$ cocompact and $\mathcal{F} = \Gamma\backslash\mathbb{H}$:

- $\Delta_{\mathcal{F}}$ self-adjoint on $L^2(\mathcal{F})$, discrete spectrum $\{\lambda_n\}$.
- Selberg zeta $Z_\Gamma(s)$: non-trivial zeros $\rho_n = 1/2 + ir_n$ with $\lambda_n = 1/4 + r_n^2$.
- Selberg-RH: $r_n \in \mathbb{R}$ automatically from $\lambda_n \geq 0$ (self-adjointness of Laplace).

The Selberg trace formula relates $\sum_n h(r_n)$ to closed geodesic data — a non-abelian Poisson-Weil-type formula.

11.3 Axiom check for Selberg

Axiom	Status	Comment
(G) abelian	no	requires non-abelian extension
(R) renormalisation	✓	geodesic flow
(N) arithmetic nucleus	partial	Tannakian analogue
(χ) character	✓	replaced by representation
(P1–P3) Pattern A	partial	spectral decomposition exists
(SGE) group structure	✓	Γ is a group
(DS1–DS3)	✓	geodesic dynamics
(RFEP)	✓	Laplace functional

Table 4: FST axiom check for the Selberg setting. The abelian axiom (G) fails; all others are *compatible* with a natural non-abelian extension.

11.4 Tannakian extension (sketch)

Replace:

- Abelian group G by Tannakian category $\mathcal{T}$ of representations.
- Character χ by irreducible representation $\pi \in \mathcal{T}$.
- Pontryagin duality by Tannakian reconstruction.

For Selberg, $\mathcal{T}$ is built from $\mathrm{Rep}(\mathrm{PSL}_2(\mathbb{R}))$ restricted to Γ-invariant submodules.

Conjecture 11.1 (non-abelian FST fixed-point theorem)**.** *Under a suitable non-abelian (Tannakian) formulation of the FST axioms and a non-abelian analogue (FST-7'), there exists an operator H_Γ on a Fréchet space $\mathcal{V}_\Gamma$ with point spectrum equal to the Selberg zero ordinates, B-symmetric for the Petersson pairing. Reality of spectrum would follow from self-adjointness (through an automatic Laplace structure), hence RH for Z_Γ.*

11.5 What the Selberg compatibility check establishes

1. **Structural compatibility.** FST (non-abelian form) is not obviously incompatible with a known RH proof — a sanity check on the axioms.
2. **Shape of a non-abelian extension.** The Tannakian vocabulary provides a natural language for non-abelian representation-theoretic content.

We do *not* claim: that the non-abelian FST axioms *re-derive* Selberg's proof, that (FST-7'(C)) follows from Anosov ergodicity in a way that is formalised as a theorem of the Tannakian FST framework, or that the non-abelian axiom package has category-theoretic maturity comparable to the abelian one. What is established here is an analogy and a compatibility test, not a proved non-abelian fixed-point theorem.

11.6 The Riemann-vs-Selberg diagnosis

The decisive difference:

- Selberg: the arithmetic-adelic analogue of (FST-7) is *geometrically natural* — the Laplace operator emerges from Riemannian metric geometry.
- Riemann: (FST-7) is *arithmetically adelic* — no analogous geometry; Meyer's Fréchet-distributional structure plays the analogous structural role.

The Bost-Connes thermodynamic mechanism (Section 13) was conjectured in earlier drafts to bridge this gap dynamically; we now regard this as open.

12 Classification via FST Structure

Family	**Type**	**SGE**	**(FST-7)**	**FST-mechanism**	**Status**
Riemann $\zeta(s)$	abelian	✓	✓ (Tate)	Meyer / CCM-prolate	open; CCM numerics strong
Dirichlet $L(s,\chi)$	abelian	✓	✓ (Tate)	Meyer-adelic	open
Hecke $L(s,\chi_K)$	abelian	✓	✓ (Tate-Weil)	Meyer-field-adelic	open
Selberg Z_Γ	non-abelian	✓	✓ (trace formula)	Laplace-direct	proved (Selberg)
Ihara-Petersen	finite	✓	N/A (finite)	trivial	proved (Bass-Hashimoto)
Artin (non-abelian)	non-abelian	✓	open	Tannakian-Langlands	open
Automorphic L on GL_n	non-abelian	✓	open	Langlands-functorial	open
$\mathbb{F}_1$-hypothetical	monoidal	?	open	Connes-Consani	open
Prime-Hub (OP5)	unclear	?	open	no natural group	open

Table 5: Classification of zeta-type families under FST axioms, with a Tannakian extension for non-abelian cases. FST here functions as an organising framework, not as a proof engine.

Observation. FST gives a predictive-looking classification: families where (FST-7) holds admit Meyer-type constructions; the non-abelian cases require a Tannakian extension (SGE still gives the group structure). Prime-Hub is unresolved because no natural group has been identified.

13 The Emergence Programme (Route C): Diagnosis of a Deep Obstruction

Early drafts of this project proposed Route C, a thermodynamic mechanism by which FST together with two additional selection axioms and Cornelissen-Marcolli rigidity would "derive" (FST-7) as a theorem. Subsequent analysis (cf. [18, 30]) shows that this route *relocates* rather than resolves the arithmetic depth. This section presents the emergence programme in its now-diagnostic form: as an identification of where the hard arithmetic input sits, not as an almost-complete path to (FST-7).

13.1 Two additional FST axioms as *selection principles*

Definition 13.1 (FST-Free axiom)**.** The multiplicative semigroup $N \leq G$ is a free commutative semigroup, generated by irreducible "primes" $\{p_i\}_{i\in I}$.

Definition 13.2 (FST-Min axiom)**.** Among all FST-structures satisfying the baseline axioms plus (FST-Free) and admitting a critical phase transition, an additional selection criterion picks out the structure whose underlying global field K is the *initial object* in the category of characteristic-zero global fields. In the abelian characteristic-zero case this selects $K = \mathbb{Q}$. This global-field language uses the classical product-formula perspective [29] as background arithmetic input.

Remark 13.3 (Honest reading)**.** (FST-Free) is a multiplicative-freedom assumption on N. (FST-Min) is a minimal-description selection criterion. Both are *strong selection principles that encode arithmetic content* — they are not consequences of the baseline axioms. In particular, in the reduction below, they function as external arithmetic imports, not as derived facts.

13.2 The Cornelissen-Marcolli bridge

Theorem 13.4 (Cornelissen-Marcolli 2010 [25])**.** *Let K_1, K_2 be number fields and let $(\mathcal{A}_{K_i}, \sigma_{K_i})$ denote their associated Bost-Connes systems. The following are equivalent:*

(i) $K_1 \cong K_2$ *as fields.*

(ii) $(\mathcal{A}_{K_1}, \sigma_{K_1}) \cong (\mathcal{A}_{K_2}, \sigma_{K_2})$ *as C^*-dynamical systems.*

In particular, the BC-system determines the underlying number field up to isomorphism.

Remark 13.5. Theorem 13.4 does *not* claim that the partition function alone determines the field (two non-isomorphic fields can share the same Dedekind zeta). The full C^*-algebra plus one-parameter group is required. This is an important hypothesis of the theorem; it is stronger than the input the emergence programme actually supplies.

13.3 Thermodynamic conditions

Following [31, 32] one extracts from FST_0 + RFEP + (FST-Free) four *thermodynamic conditions* on the associated QSM system $(\mathcal{A}, \sigma)$:

(B1)

Dirichlet structure. $Z(\beta) = \sum_{n\in X} w_n\, n^{-\beta}$ for a countable index set X and positive weights w_n.

(B2)

Euler product. $Z(\beta) = \prod_{p\in P}(1 - w_p p^{-\beta})^{-1}$ for an atomic generator set P ("primes").

(B3)

Simple pole at $\beta = 1$. $\lim_{\beta\to 1^+}(\beta - 1)\, Z(\beta) = c > 0$.

(B4)

FST_0-conformity. $\mathcal{A}$ is the universal Pattern-A crossed product of N with its one-parameter scaling.

Proposition 13.6 (Thermodynamic derivation, partial)**.** *Under FST_0 baseline axioms together with (FST-Free) and RFEP with a one-parameter thermodynamic variable, there is a heuristic route to (B1), (B2), and (B4); (B3) requires a further arithmetic input (identification of the thermodynamic pole with the Dedekind pole).*

Remark 13.7 (Status of Proposition 13.6)**.** The proof sketches in v0.4–v0.6 presented this as a straightforward derivation. The honest position is: (B1), (B2), (B4) follow from RFEP + (FST-Free) by standard critical-exponent style arguments; (B3) — which is the content with arithmetic bite — is where the identification with a genuine number-field Dedekind pole enters. This does not follow from the thermodynamic side alone; it requires the input developed in the gap analysis of [18].

13.4 The former "main reduction theorem"

Earlier versions of this paper stated a Theorem ("Reduction of the Emergence Hypothesis") asserting that FST_0 + (FST-Free) + (FST-Min) + RFEP + Cornelissen-Marcolli rigidity force the underlying QSM system to be the BC system of $\mathbb{Q}$, and hence entail (FST-7)(A) via Tate. The updated position is that this chain has a crucial open step at (B3), and more fundamentally that the identification of a QSM system $(\mathcal{A}, \sigma)$ satisfying (B1)–(B4) with the BC system of a *number field* K requires *more* than a Dirichlet/Euler/pole pattern: it requires the reconstruction of the additive structure of K from data that is prima facie only multiplicative.

We state this formally:

Conjecture 13.8 (L1-K: reconstruction of additive structure)**.** *A QSM system* $(\mathcal{A}, \sigma)$ *satisfying (B1)–(B4) plus (FST-Min) is isomorphic as a* C^**-dynamical system to the Bost-Connes system of a number field* $K = \mathbb{Q}$*.*

Remark 13.9 (Why L1-K is hard)**.** Cornelissen-Marcolli characterise a number field through its BC system, but the hypotheses of that theorem include the full C^*-dynamical structure of a BC system — in particular the pieces encoding the ring structure of the ring of integers, including addition; see also the number-field Toeplitz-type construction of Cuntz-Deninger-Laca [28]. The thermodynamic conditions (B1)–(B4) are primarily multiplicative. Closing this gap is classically equivalent to imposing an additive structure compatible with the multiplicative one, i.e. the additive-multiplicative coupling problem addressed by the Connes-Consani arithmetic-site / $\mathbb{F}_1$ programme [23]. No version of the emergence programme developed in this project resolves this gap. Within our formal language this corresponds to introducing yet another axiom, *(FST-Add)*, which encodes the additive structure of K — in which case the argument becomes transparently circular: we are assuming what we want to derive.

13.5 Blueprint for Route C, in diagnostic form

Conditional Blueprint 13.10 (Route C, in diagnostic form)**.** *Under* FST_0 *+ (FST-Free) + (FST-Min) + RFEP + (B3),* and conditional on Conjecture 13.8 (L1-K)*, the chain*

$$\textit{FST data} \;\to\; \textit{QSM system satisfying (B1)–(B4)} \xrightarrow{\textit{CM rigidity}} \textit{BC system of } \mathbb{Q} \xrightarrow{\textit{Tate}} \textit{(FST-7)(A)}$$

closes. In this scenario (FST-7) would be a theorem. In practice, Conjecture 13.8 is the additive-multiplicative coupling problem of the $\mathbb{F}_1$ *programme; the emergence programme therefore does* not *close the loop autonomously.*

13.6 Demotion of Route C

Remark 13.11 (Demotion)**.** Consequently we demote Route C from a candidate "main programme" to a diagnostic side investigation. Its contribution is structural clarification: by following the emergence chain one can identify the additive-multiplicative coupling as the single deep obstruction, and one can see explicitly that BC-thermodynamics alone does not supply additive structure. This is useful as a diagnostic result about where the arithmetic depth lives; it is *not* useful as a proof engine.

13.7 The three parallel routes under the FST umbrella

With Route C demoted, the route picture is:

Route	Mechanism	(FST-7) view	Key object	Status
(A) Meyer-Tate	adelic self-duality	Pontryagin	$\mathbb{A}_{\mathbb{Q}}$, $\mathbb{Q}$-lattice	context; (O1)–(O3) open
(B) CCM-prolate	time-frequency optimality	Nyquist-Shannon	$D_{\log}^{(\lambda,N)}$, PW_λ	**main program; 4 milestones**
(C) BC-emergence	thermodynamic phase transition	arithmetic ergodic	BC-algebra	demoted (L1-K = add/mult coupling)

By Schema 7.7 the three views of (FST-7) are compatible structural readings of the same supplied input, so each route, if completed with its own missing hypotheses, would entail (FST-7) and hence (in combination with the Meyer construction or its analogue) the Riemann Hypothesis. The point is that completion means very different things on the three routes, and at present only Route B is pursued as an active technical programme.

Remark 13.12 (Role of FST in v1.8)**.** FST does not compete with any of the three routes. It provides a common *classification and comparison frame*: the same (FST-7) axiom, in three compatible structural forms, is attacked by three different technical programmes. The deep content is the structural *compatibility* — that adelic self-duality, time-frequency optimality, and arithmetic ergodicity are three structural faces of a single phenomenon. FST does not, by itself, prove anything about RH.

14 Open Research Questions

(OFQ1) **(FST-7) from RFEP.** Open; the emergence route identifies (L1-K), the additive-multiplicative coupling, as the core obstruction (Section 13).

(OFQ2) **Resolution of (O1)–(O3).** Technical completion of the Wick-Mellin rotation; (O3) is the hardest piece and its trace-class status is open (Conjecture 10.3).

(OFQ3) **Non-abelian FST.** Tannakian axioms as a vocabulary for Artin and automorphic L-functions; Conjecture 11.1 open; cf. [33] for exploratory formalisation.

(OFQ4) **Selberg compatibility.** Compatibility established (Section 11); a full non-abelian fixed-point theorem remains open.

(OFQ5) **$\mathbb{F}_1$-FST and the additive-multiplicative coupling.** The (L1-K) obstruction of Section 13 coincides with the classical Connes-Consani arithmetic-site programme; this is the long-term frame in which Route C could conceivably be resurrected.

(OFQ6) **Physical realisation.** Bost-Connes in condensed matter; experimental phase transition (speculative).

(OFQ7) **Rigidity.** Adelic structure: Margulis-type rigidity or moduli? (exploratory).

15 Status, Honest Assessment, and Outlook

15.1 What this paper establishes

1. Axiomatic FST framework for spectrum-zero identification, with explicit separation of baseline and completeness axioms (Sections 3, 5).
2. Identification of (FST-7) with three structurally compatible forms (Section 7); the three forms are, in practice, three distinct technical entry points, not interchangeable proof strategies.
3. Three technical obstacles (O1)–(O3) on Route A, catalogued with candidate construction sketches but *not* reduced to standard techniques (Section 10).

4. Selberg compatibility check: a non-abelian Tannakian extension is compatible with Selberg's RH proof (Section 11); it does not re-derive the proof.
5. Classification of zeta-type families (Table 5).
6. A **Route B reduction blueprint** for the CCM missing-step language (Blueprint 9.6), together with a four-milestone agenda for Route B (Milestones 9.9–9.12).
7. **(New in v0.8; promoted by Zookeeper)** A **Three-Lemma Endgame** (Section 9.7) that reduces (MS2/C2) in the Zookeeper paper to a conditional mass-based microcluster architecture with the uniform gates $s_\lambda \to 0$, $p_\lambda \to 0$, and $g_* \geq g_0 > 0$ still explicit: three separate conditional inputs (Lemmas 9.17–9.19) yield the mass concentration bound (Theorem 9.20) as a corollary of a standard quasimode-to-projector inequality.
8. **(New in v0.8)** An RFEP transfer package for physical supplements (Section 9.8), translating the Zookeeper operator pattern into explicit proof obligations: operator/form, defect, Galerkin faithfulness, complement gap, and domain limit.
9. **(New in v1.8)** A branch-local domain-status table (Table 1) separating RFEP as physical meta-rule from the Zeta Zoo as mathematical companion branch.
10. **(New in v1.10 candidate revision)** A tangential certificate-ledger table (Table 3) that separates raw total residuals from projected tangent evidence and keeps mechanical positive controls below domain-certificate level.
11. A diagnostic reformulation of Route C identifying the additive-multiplicative coupling (L1-K) as its core obstruction (Section 13).

15.2 What this paper does not establish

1. No unconditional new theorem about RH or GRH.
2. No derivation of (FST-7) from the baseline FST axioms alone; only a conjectural *relocation* of the arithmetic input under additional selection axioms.
3. No resolution of the Wick-Mellin obstacles (O1)–(O3); (O3)'s trace-class claim from v0.6 is explicitly retracted (Remark 10.4).
4. No unconditional proof of (MS1) (simple-even for QW_λ); this remains open.
5. No independent proof of the Zookeeper MS2/C2 uniform constants inside this RFEP paper. This paper imports the Zookeeper microcluster architecture as a conditional normal-form template for physical transfer.
6. No automatic transfer of the Zookeeper architecture to physical supplements. Each physical branch must still verify its own analogue of the operator/form, defect, quasimode, complement gap, and domain-limit hypotheses.
7. No proof of a generalised Weyl asymptotic for QW_λ; replaced by Milestone 9.12.
8. No proof of the physical supplements from the RFEP transfer schema. Proposition 9.23 is a reusable conditional template; Yang–Mills, Navier–Stokes, NS-LDI, turbulence, and CRM must verify its hypotheses separately.
9. No physical-domain row currently passes the full v1.10 candidate tangential-provenance ledger. Existing positive rows are mechanical controls for the bookkeeping rule, not certificates for Yang–Mills, Navier–Stokes, NS-LDI, turbulence, or CRM.
10. No proof of the non-abelian fixed-point theorem (Conjecture 11.1).

11. No proof of Proposition 13.6 for (B3); no proof of Conjecture 13.8 (L1-K); no resolution of the additive-multiplicative coupling.

15.3 Honest assessment

FST in its present form is a comparative framework with a Zookeeper-conditioned C2/MS2 microcluster normal form. It provides structural clarity, a classification criterion, and an explicit three-route picture in which the hard open content is localised:

- On Route A, the hard content sits in the Wick-Mellin obstacles (O2), (O3).
- On Route B, the Zookeeper paper reduces the approximation problem (MS2/C2) to the Three-Lemma Microcluster architecture plus uniform control of s_λ, p_λ, and g_*. In this RFEP paper, that conditional architecture is imported rather than reproved; its role is to supply the normal form for physical transfer.
- On Route C, the hard content sits in Conjecture 13.8 (L1-K), which is the classical additive-multiplicative coupling problem and is outside the reach of the present project.

This RFEP paper does not independently prove RH. It records the post-Zookeeper status: Route B's approximation problem (MS2/C2), previously identified as a deep open item in the CCM programme, has been reduced to explicit microcluster estimates and uniform-constant gates in the Zookeeper paper. The RFEP consequence is structural: the physical branches inherit a precise normal form, not an automatic solution of their own PDE or continuum-limit problems.

For the physical branch, the gain is different in kind: RFEP now has a transfer ledger rather than a slogan. Physical supplements can inherit the Zookeeper pattern only after they identify their operator normal form, defect, Galerkin limit, complement gap, and tangential-versus-normal residual provenance.

15.4 Role of the framework

- *Uniform language* for Tate, Meyer, Bost-Connes, Selberg, CCM, and Connes-Consani.
- *Classification criterion* (Table 5).
- *Research agenda* (OFQ1–OFQ7; milestones 9.9–9.12 for Route B).
- *Zookeeper conditional microcluster reduction* for Route B's (MS2/C2) via the Three-Lemma architecture.
- *RFEP transfer package* for physical supplements (Section 9.8).
- *Diagnostic identification* that Route C's gap is the additive-multiplicative coupling.
- *Compatibility sketch* with Selberg (Section 11).

15.5 Next technical steps

1. **Route B, Milestone 9.9:** Domain/form theory for Ψ_λ as a bounded intertwiner between suitable form domains. This is the first rigorous step and is the prerequisite for any precise statement of Blueprint 9.6.
2. **Route B, Milestone 9.10:** min-max low-eigenvalue comparison, not full Weyl asymptotic.
3. **Route B, Milestone 9.11:** finite-dimensional CCM truncation theorem inside E_N.
4. **Route B, Milestone 9.12:** only after 1–3, asymptotic regime.

5. **RFEP physical transfer:** replace the five-field checklist by a tangential certificate ledger. Each supplement records, in this order, target predefinition, source-axis independence, independent outer gap, Galerkin exactness or no-pollution status, total residual/gap, projected-residual/tangent-gap, normal residual, whether the normal part is paid by explicit constraints, active constraints, projection predefinition, sloppy/null-direction class, scale degree, advice source, matched controls, and final diagnostic/certificate decision. Prime-Hub remains the first natural test case because its frontier-prime defects already have the right finite-rank/boundary flavour.
6. **Route A:** operator-theoretic analysis of (O3) and precise articulation of the limiting KMS states required by (O2); the trace-class question (Conjecture 10.3) is open.
7. **Route C:** documentation only; no further attack without progress on additive-multiplicative coupling.
8. **Non-abelian (OFQ3):** Langlands functoriality may be a better long-term frame than the present Tannakian FST formulation; cf. [33].
9. **Axiomatics:** independence/minimality proofs for the baseline axioms; separation of setup, dynamical, and arithmetic-completeness assumptions (Remark 3.1).

Each of these is a multi-month research task.

15.6 The CRM branch

The Cooperative Renormalization Model (CRM) places the cosmology branch inside the RFEP v1.10 candidate map. In the mathematics branch, constrained positivity and self-duality organise spectral candidates. In the physics branch, DS1–DS3 describe how dissipative dynamics selects a stable attractor. In the cosmology branch, CRM is the proposed large-scale instantiation: cooperative renormalisation replaces a single local defect by a coupled background response, and RFEP requires the observed dark-energy behaviour to appear as a stable minimum of the corresponding renormalised free-energy functional.

The CRM claim is therefore intentionally weaker than an observational discovery claim. The present paper records the structural placement: CRM is the cosmological RFEP branch whose remaining obligations are model identification, parameter stability, and observational discriminants. Those tasks belong to the dedicated CRM/cosmology paper, not to the Zeta Zoo proof branch.

15.7 Relation to the Math-Master

The present paper's Tannakian extension (Section 11) refines the Math-Master HP-BL classification [1], Sections 2.4 and 9: even on the NO side (Riemann, Dirichlet), an FST-spectral realisation may exist via the idele-side SGE-YES route. The three-component decomposition $\mathrm{RH}(X) \iff \mathrm{GBC}(X) \wedge C^{(2)}(X) \wedge \mathrm{Hurwitz}(X)$ of the Math-Master gets its $C^{(2)}$ component identified with the Meyer construction (as discussed in [7]). This identification is structural, not a proof of the Math-Master decomposition's implications.

Provenance Note

This paper is a position draft prepared within the META_RH_TREE project of FST-Mathematics. Relative to v0.6, it is an honest rewrite that incorporates the April 2026 external audit (`_audit/GPT_REVIEW_2026-04-17_*.md`) and brings paper-level claims into line with the honest note-level diagnoses. It consolidates the Session-11 to Session-16 explorations documented in the internal project folder `CORE/RFEP/`, including concept, obstruction, bridge, derivation, and Zookeeper-transfer notes. These notes are workflow provenance and internal audit material; the public scientific claims are the claims stated and delimited in this manuscript.

The July 2026 v1.10 candidate revision adds the internal Tangential-Provenance-Ledger audit to the manuscript-level claim contract. It changes no theorem statement and promotes no physical branch claim; it only makes explicit that projected tangent evidence, paid normal residuals, independent gap provenance, and matched controls are prerequisites for any future certificate-level transfer.

AI Disclosure / KI-Offenlegung

Erklärung zum Einsatz von KI-Werkzeugen

Dieses Manuskript wurde unter umfassender Unterstützung durch KI-gestützte Sprachmodelle (Claude, Anthropic; Gemini, Google; ChatGPT, OpenAI) erstellt. Die KI-Werkzeuge wurden in mehreren Phasen des Forschungs- und Schreibprozesses eingesetzt, darunter konzeptionelle Diskussion, Literaturrecherche, Strukturierung, Formulierungshilfe, Analyse- und Synthesevorschläge, kritische Gegenprüfung, sprachliche Überarbeitung, Übersetzung, Zitations- und Quellenarbeit sowie technische LaTeX-Formatierung, Satzvorbereitung und Kompilierungsunterstützung.

Für rechnerische, mathematische oder simulationsbezogene Bestandteile dieser Arbeit wurden KI-Werkzeuge zusätzlich zur Erstellung, Anpassung, Umschreibung, Dokumentation und Prüfung von Python-Skripten sowie zur Planung, Durchführung und Betreuung von Berechnungsläufen eingesetzt. Dies umfasste insbesondere die Weiterentwicklung bestehender Skripte, die Vorbereitung nachvollziehbarer Ausgaben, die Auswertung von Zwischenergebnissen, Reanalysen und die Dokumentation von Tests, Verfahren und Ergebnissen.

Zur Qualitätssicherung wurden, soweit einschlägig, mehrere komplementäre Verfahren eingesetzt: Modell-Cross-Checks, mehrstufige Review-Pipelines, explizit widerlegungsorientierte Prüfungen, Quellen- und Zitationschecks, sprachliche Gegenlesungen, Reanalysen, reproduzierbare Skripte, gespeicherte Resultatdateien, Testläufe, Hash- oder Toleranzvergleiche, erneute Ausführungen sowie eine ordnerbasierte Forschungspipeline mit Projekt-Guidelines, vorgegebenen Workflows, Dokumentationsregeln und dokumentierten Korrekturschritten.

Modell- und Workflow-Zuordnung.

In der bisherigen Forschungsarbeit wurden einfachere Aufgaben typischerweise mit Opus/Sonnet 4.6, GPT-5.4 und Gemini 3.1 Pro bearbeitet. Gemini 3.1 Pro wurde insbesondere auch für Übersetzungen eingesetzt. Anspruchsvollere Aufgaben wurden mit Opus 4.6, Opus 4.7 und GPT-5.5 durchgeführt. Opus 4.6 wurde häufig als dialogischer Forschungspartner zur Entwicklung und Fortführung des Forschungsprozesses eingesetzt, während Opus 4.7 und GPT-5.5 verstärkt für Reviews, Audits, Gegenprüfungen und kritische Nachanalysen genutzt wurden. GPT-5.5 wurde insbesondere bei mathematischen Fragestellungen eingesetzt. Alle genannten Modellfamilien wurden, soweit einschlägig, auch für LaTeX-Satz, technische Formatierung, Kompilierung und Fehlerdiagnose eingesetzt. Methodisch kamen iterative Schleifen, Modell-Cross-Checks, Interchat-Gespräche zwischen Opus-, Gemini- und GPT-basierten Systemen sowie ein mehrstufiges 7-Schritte-Review-Verfahren mit explizitem Widerlegungsagenten und anschließendem Heilungs- bzw. Reparaturversuch zum Einsatz. Der Mensch fungierte als Gedächtnis, verbindendes und antreibendes Element, reflektierende Instanz und Orchestrator des Gesamtprozesses; er brachte zudem häufig Lösungen gerade durch den Versuch ein, die offenen Probleme selbst zu verstehen.

Der Autor bestimmte Forschungsfrage, methodische Ausrichtung, zentrale Thesen, Bewertung der Quellen, fachliche Interpretation und Schlussfolgerungen.

Statement on the Use of AI Tools

This manuscript was prepared with extensive assistance from AI-based language models (Claude, Anthropic; Gemini, Google; ChatGPT, OpenAI). AI tools were used across multiple stages of the research and writing process, including conceptual discussion, literature search, structuring, drafting support, suggestions for analysis and synthesis, critical counter-checking, language revision, translation, citation and source work, technical formatting, LaTeX typesetting, and compilation support.

For computational, mathematical, or simulation-related components of this work, AI tools were also used to create, adapt, rewrite, document, and check Python scripts, and to support the planning, execution, and supervision of computational runs. This included the further development of existing scripts, the preparation of traceable outputs, the evaluation of intermediate results, reanalyses, and the documentation of tests, procedures, and results.

As part of quality assurance, where applicable, several complementary procedures were used: model cross-checks, multi-stage review pipelines, explicitly falsification-oriented reviews, source and citation checks, linguistic cross-reading, reanalyses, reproducible scripts, stored result files, test runs, hash or tolerance comparisons, repeated executions, and a folder-based research pipeline with project guidelines, predefined workflows, documentation rules, and documented correction steps.

Model and Workflow Assignment.

In the research work to date, simpler tasks were typically handled with Opus/Sonnet 4.6, GPT-5.4, and Gemini 3.1 Pro. Gemini 3.1 Pro was also used in particular for translation work. More demanding tasks were carried out with Opus 4.6, Opus 4.7, and GPT-5.5. Opus 4.6 was often used as a dialogic research partner for developing and advancing the research process, whereas Opus 4.7 and GPT-5.5 were used more prominently for reviews, audits, counter-checks, and critical reanalyses. GPT-5.5 was used in particular for mathematical questions. Where applicable, all listed model families were also used for LaTeX typesetting, technical formatting, compilation, and error diagnosis. Methodologically, the work involved iterative loops, model cross-checks, inter-chat exchanges between Opus-, Gemini-, and GPT-based systems, and a multi-stage seven-step review procedure with an explicit falsification agent followed by a repair or reconstruction attempt. The human author served as the memory, connective and driving element, reflective instance, and orchestrator of the overall process; he also often contributed solutions through the attempt to understand the open problems himself.

The author determined the research question, methodological direction, central claims, evaluation of sources, disciplinary interpretation, and conclusions.

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